

# Computational Viability Certification under Incomplete Observation

## Supplementary Material: Complete Proofs and the Verification Record

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### S1. Complete proofs

Throughout, labels are  $a = (\theta, i, t) \in \mathcal{S} = \Theta \times \{1, \dots, s\} \times [0, T]$ ,  $k_a$ ,  $\beta_a$ ,  $\mathcal{F}_a$ ,  $J_{\mathcal{I}}(T)$ ,  $\phi_U$ ,  $\Gamma_{\mathcal{I}}$ ,  $\rho_{\mathcal{H},G}$ , and the program (5) are as in the main text.

**Proof of Proposition 1 (robust rows).** The linear solution formula gives, for  $t \leq T$ ,

$$c_i^\top x_\theta(t) = c_i^\top X_\theta(t, 0)x_0 + \int_0^t c_i^\top X_\theta(t, \sigma)g_\theta(\sigma) d\sigma + \int_0^t k_a(\sigma)^\top u(\sigma) d\sigma + \int_0^t c_i^\top X_\theta(t, \sigma)E_\theta(\sigma)d(\sigma) d\sigma.$$

The initial-state term is bounded over  $x_0 \in P_\theta$  by the support value  $h_{P_\theta}(X_\theta(t, 0)^\top c_i)$ , and the disturbance term, maximized over all measurable  $d(\cdot)$  with values in  $D_\theta$ , is  $\int_0^t h_{D_\theta}(E_\theta(\sigma)^\top X_\theta(t, \sigma)^\top c_i) d\sigma$ ; for polytopic  $D_\theta$  a maximizing vertex selection is measurable, so the bound is attained by an admissible disturbance signal. Collecting terms, the constraint  $c_i^\top x_\theta(t) \leq b_i$  holds for every  $(x_0, d)$  if and only if  $\mathcal{F}_a(\pi) \leq 0$ , and the separation of the maximizations from the control term is exactly that the control signal in a mode does not depend on  $x_0$  or  $d$ .

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**Proof of Proposition 2 (value and nonviability).** The admissible signal space is a finite product of the sets  $L^\infty(I_r; U)$ ; because  $U$  is compact and convex these are weak-\* compact by the Banach–Alaoglu theorem. The safety functionals are affine and weak-\* continuous because their kernels  $k_a$  lie in  $L^1$ . Their supremum over the finitely many labels is therefore weak-\* lower semicontinuous and attains its minimum on the policy space:  $J_{\mathcal{I}}(T)$  is well defined. Safety of some policy is  $\max_a \mathcal{F}_a(\pi) \leq 0$ , so  $B_0 \in \mathcal{K}_{\mathcal{I}}^T$  iff  $J_{\mathcal{I}}(T) \leq 0$ ; since safety at the margin is viability (constraints closed) and  $J_{\mathcal{I}}(T) = 0$  admits safe policies, nonviability is  $J_{\mathcal{I}}(T) > 0$ . The strict inequality is thus a consequence of compactness and closedness, not an added assumption.

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**Proof of Theorem 4 (continuous-to-finite).** *Step 1 (control-moment inequality).* For a block  $J_j$  and any measurable  $u$  with values in  $U$ ,

$$\left(\int_{J_j} k_a\right)^\top \frac{1}{|J_j|} \int_{J_j} u - \int_{J_j} k_a^\top u = \int_{J_j} (\bar{k}_{a,\mathcal{H}} - k_a)^\top u \leq \int_{J_j} h_U(\bar{k}_{a,\mathcal{H}} - k_a),$$

using  $\sup_{v \in U} w^\top v = h_U(w)$ . Summing over blocks proves (3); the averages remain compatible with the information structure because the mesh contains all sensing times, so common-history controls use the same block variable. For the bound, choose a center  $u_c$  of  $U$ ; since  $\int_{J_j} (k - \bar{k}) = 0$ , the center cancels and  $e_{a,\mathcal{H}} \leq R_U \|k_a - \bar{k}_{a,\mathcal{H}}\|_{L^1}$ . A bounded-variation function satisfies  $\|k_a - \bar{k}_{a,\mathcal{H}}\|_{L^1} \leq h_u \text{TV}(k_a)$ , whence  $e_{a,\mathcal{H}} \leq R_U V_U h_u$ . The pointwise bound behind it: for every  $\sigma \in J$ ,  $\|\bar{k} - k(\sigma)\| = \left\| \frac{1}{|J|} \int_J (k(\tau) - k(\sigma)) d\tau \right\| \leq \text{TV}(k; J)$ , the norm of any increment being at most the total variation; integrate over  $J$  and sum. The same estimate bounds the reflected integrand  $\int h_U(k_a - \bar{k})$  (center cancellation is symmetric), which is what Step 3 uses.

*Step 2 (lower bound of (8)).* Take any admissible continuous-time policy and form its block averages, which lie in  $U$  and satisfy (3). For every grid label,

$$A_a v - \beta_a^+ - \bar{e}_a \leq \int_0^T k_a^\top u_\theta^\pi - \beta_a = \mathcal{F}_a(\pi),$$

so the finite program attains a value no larger than the maximum robust violation of that policy (the display needs only the one-sided bound from (3),  $A_a v - \int k_a^\top u^\pi \leq \bar{e}_a$ , and  $\beta_a^+ \geq \beta_a$ . The reflected error can be larger than  $\bar{e}_a$ , so an absolute-value bound is not inferred). Minimizing over policies gives  $\rho_{\mathcal{H},G} \leq J_{\mathcal{I}}(T)$  — the block-moment vector of every admissible policy is feasible for the finite program, at objective  $\max_a \mathcal{F}_a(\pi)$ .

*Step 3 (upper bound of (8)).* Take an optimizer  $v^*$  of (5) and implement the corresponding piecewise-constant information-adapted control, for which  $\int_0^T k_a^\top u_\theta^{v^*} = A_a v^*$  — exactly, with no moment error, because the implemented control is constant on each block; the implementation price is paid only through the grid labels

versus all times below. At grid times,  $\mathcal{F}_a(v^*) \leq \rho_{\mathcal{H},G} + \bar{e}_a + \delta_\beta$ ; the Lipschitz hypothesis (6) extends the bound to every time ( $|\mathcal{F}_a(v^*) - \mathcal{F}_{\hat{a}}(v^*)| \leq L|t - \hat{t}| \leq Lh_t$  for any time label within  $h_t$  of the grid label  $\hat{a}$ ), giving  $\max_a \mathcal{F}_a(v^*) \leq \rho_{\mathcal{H},G} + \max_{\hat{a} \in \mathcal{S}_G} \bar{e}_{\hat{a}} + \delta_\beta + Lh_t$ . Since  $J_{\mathcal{I}}(T)$  minimizes over all admissible policies, the upper bound follows. Together with (4) this gives the sharpened form; the uniform choice  $\bar{e} \equiv R_U V_U h_u$  — valid by the moment-error bound (2) whenever  $\text{TV}(k_a) \leq V_U$  — recovers the announced special case. The asymmetry — averaging arbitrary controls proves the lower bound, implementing piecewise-constant controls proves the upper bound — is what makes naïve infeasibility-by-discretization arguments unsound.

*Step 4 (sound finite certificates).* Write the block input constraints as  $F_{\text{blk}} v \leq f_{\text{blk}}$ . The program's dual is

$$\max_{\lambda, \mu} -\lambda^\top (\beta^+ + \bar{e}) - \mu^\top f_{\text{blk}} \quad \text{s.t.} \quad A^\top \lambda + F_{\text{blk}}^\top \mu = 0, \quad \lambda \geq 0, \quad \mathbf{1}^\top \lambda = 1, \quad \mu \geq 0, \quad (12)$$

so a positive dual objective certifies infeasibility of  $\{Av \leq \beta^+ + \bar{e}, v \in U^Q\}$ . By Step 1 every genuinely safe measurable controller would generate a feasible  $v$ ; the finite infeasibility certificate therefore excludes every such controller. Moreover the weighted continuous-time violation is at least the dual objective: a normalized certificate with margin  $\gamma > 0$  guarantees an actual facet violation of at least  $\gamma$  in one stored test (Remark 1 of the main text).

*Step 5 (completeness).* If the system is nonviable, let  $\gamma = J_{\mathcal{I}}(T) > 0$  and choose meshes and stored scenarios making the error in (8) — including the stored-scenario widths  $\delta_{\text{init}}$  and  $\delta_{\text{num}}$ , which near-maximizing stores contribute and completeness must drive to zero — smaller than  $\gamma$ ; then  $\rho_{\mathcal{H},G} > 0$  and Step 4's finite LP duality yields a positive certificate, proving (iii). No belief-state recursion is used.

*Step 5b (exact atomic dual representation, proof of (9)).* Write  $\Pi$  for the policy space and  $\mathcal{S}$  for the label set.

(a) *Policy compactness.* Each factor  $\{u \in L^\infty(I_r; U)\}$  is weak-\* closed —  $U$  is a finite intersection of halfspaces  $f_i^\top v \leq g_i$ , and for  $f_i \mathbf{1}_{I_r} \in L^1$  the set  $\{u : f_i^\top u(\sigma) \leq g_i \text{ a.e.}\}$  is weak-\* closed — and weak-\* bounded; Banach–Alaoglu makes it weak-\* compact and convex. The finite product  $\Pi$  is therefore weak-\* compact, convex, and metrizable (each  $L^1$  is separable, so bounded weak-\* sets are metrizable).

(b) *Joint continuity of the label-policy pairing.* Each  $\mathcal{F}_a$  is a finite sum of  $L^1$ -pairings against cell signals minus a constant, so  $\pi \mapsto \mathcal{F}_a(\pi)$  is weak-\* continuous. Under the label-continuity hypothesis ( $\|k_a - k_{a'}\|_{L^1} \rightarrow 0$ ,  $|\beta_a - \beta_{a'}| \rightarrow 0$ ; the  $L^1$  form is the operative one — the sup-norm form is false for the step kernels, which jump at  $\sigma = t$  for every  $t$ ),

$$\sup_{\pi} |\mathcal{F}_a(\pi) - \mathcal{F}_{a'}(\pi)| \leq M_U \|k_a - k_{a'}\|_{L^1} + |\beta_a - \beta_{a'}| \rightarrow 0,$$

with  $M_U := \sup_{v \in U} \|v\|$  (Hölder: the policies take values in  $U$ , so each pairing against an  $L^1$  kernel difference is bounded by  $M_U$  times its  $L^1$  norm), uniformly in  $\pi$ ; with separate continuity this gives joint continuity of  $(a, \pi) \mapsto \mathcal{F}_a(\pi)$  on the compact  $\mathcal{S} \times \Pi$ , in particular boundedness.

(c) *Mixed extension and Sion.* Let  $\mathcal{P}(\mathcal{S})$  be the Borel probability measures with the weak-\* topology: by the Riesz representation theorem it is compact, convex, and metrizable. The bilinear  $G(\pi, \mu) = \int \mathcal{F}_a(\pi) d\mu(a)$  is affine in each variable — hence simultaneously quasi-convex and quasi-concave — and continuous in each variable ( $\pi$ -continuity by dominated convergence against the fixed finite  $\mu$ ;  $\mu$ -continuity by pairing against the continuous  $a \mapsto \mathcal{F}_a(\pi)$ ). Sion's minimax theorem (both spaces compact convex) gives  $\min_{\pi} \max_{\mu} G = \max_{\mu} \min_{\pi} G$  with the outer maximum attained; and  $\max_{\mu} G(\pi, \cdot) = \max_a \mathcal{F}_a(\pi)$ , because the integral of a continuous function over a probability measure lies within its min–max range while the Dirac measures attain both ends. Hence  $J_{\mathcal{I}}(T) = \max_{\mu} \min_{\pi} G$ .

(d) *Continuity of the value function.*  $g(\mu) := \min_{\pi} G(\pi, \mu)$  is weak-\* continuous: by (b) the family  $\{a \mapsto \mathcal{F}_a(\pi) : \pi \in \Pi\}$  is uniformly bounded and uniformly equicontinuous in  $C(\mathcal{S})$ , so Arzelà–Ascoli makes its closure compact, and  $\mu_n \rightarrow \mu$  weak-\* forces  $\sup_{\pi} |G(\pi, \mu_n) - G(\pi, \mu)| \rightarrow 0$ ; evaluating at near-minimizing  $\pi$ 's gives  $g(\mu_n) \rightarrow g(\mu)$ .

(e) *Atomic approximation.* Finitely supported measures are weak-\* dense in  $\mathcal{P}(\mathcal{S})$ : quantize  $\mathcal{S}$  into finitely many Borel sets of diameter below  $1/n$ , keep one representative point per set, and give it the set's  $\mu$ -mass; the resulting finitely supported measures converge to  $\mu$ . Taking  $\mu_n \rightarrow \mu^*$ , the maximizer of (c), (d) gives  $g(\mu_n) \rightarrow g(\mu^*) = J_{\mathcal{I}}(T)$ . Hence

$$J_{\mathcal{I}}(T) = \sup \left\{ \min_{\pi} \sum_{\ell} \lambda_{\ell} \mathcal{F}_{a_{\ell}}(\pi) : q \in \mathbb{N}, a_{\ell} \in \mathcal{S}, \lambda \in \Delta_q \right\},$$

which is (9) once the inner minimum is identified with  $\Gamma_{\mathcal{I}}$ .

(f) *The  $\Gamma$  representation.* For  $\mu = \sum_{\ell} \lambda_{\ell} \delta_{a_{\ell}}$  the inner minimum decomposes over information cells  $(r, C)$ : it is the infimum over  $u_{r,C} \in L^\infty(I_r; U)$  of  $\int_{I_r} g_{r,C}(\sigma)^\top u_{r,C}(\sigma) d\sigma$  with  $g_{r,C} = \sum_{\ell: \theta_{\ell} \in C} \lambda_{\ell} k_{a_{\ell}}$ . The inequality  $\leq \int \phi_U(g_{r,C})$  is the pointwise bound; the converse is measurable selection — for a.e.  $\sigma$  the set of minimizers of  $v \mapsto g_{r,C}(\sigma)^\top v$  over the polytope  $U$  is a nonempty face, the lexicographically first minimizing vertex in a fixed

vertex order is measurable in  $\sigma$ , and selecting it attains the pointwise minimum a.e., so the infimum equals  $\int_{I_r} \phi_U(g_{r,C})$ . Summed over cells,  $\min_\pi \sum_\ell \lambda_\ell \mathcal{F}_{a_\ell}(\pi) = \Gamma_{\mathcal{I}}(\lambda, a)$  as defined in (1).

(g) *Nonviability equivalence.* If  $J_{\mathcal{I}}(T) > 0$ , (e) supplies a finitely supported  $\mu$  with  $g(\mu) > J_{\mathcal{I}}(T)/2 > 0$ : a positive finite atomic certificate. Conversely  $\Gamma_{\mathcal{I}}(\lambda, a) = \min_\pi \sum_\ell \lambda_\ell \mathcal{F}_{a_\ell}(\pi) \leq \min_\pi \max_a \mathcal{F}_a(\pi) = J_{\mathcal{I}}(T)$  because the maximum bounds the  $\lambda$ -weighted average.  $\square$

*Step 6 (sparsity).* Factor the finite row map through its rank,  $Av = L(Rv)$ ,  $Rv \in \mathbb{R}^r$ ; the attainable moment set  $\mathcal{M} = \{Rv : v \in U^Q\}$  is compact and convex, and each safety row defines a halfspace in  $\mathbb{R}^r$ . If the common feasible set is empty, Helly's theorem in  $\mathbb{R}^r$  gives an obstruction involving at most  $r + 1$  safety halfspaces, and finite duality on that subproblem yields a sparse positive certificate with the same objective:  $r + 1 \leq mQ + 1$  labels. To preserve a prescribed margin  $\gamma_0 < \rho_{\mathcal{H},G}$ , apply the argument to the still-infeasible shifted constraints  $A_a v \leq \beta_a^+ + \bar{e}_a + \gamma_0$ . The continuous finite-rank case follows identically on the compact attainable moment body of the lifted kernels, giving  $r_{\mathcal{I}} + 1$ .  $\square$

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**Proof of Proposition 3 (arbitrarily large minimal obstructions).** With the construction of the main text and  $w_j = \int_{I_j} u$ : mode  $j \leq q$  ends the interval at  $x_j = w_j - 1 \geq 0$ , so safety of the blind window requires  $w_j \geq 1$ ; mode  $q + 1$  ends at  $q - \varepsilon - \sum_j w_j$ , so safety requires  $\sum_j w_j \leq q - \varepsilon$ . Summing the  $q$  lower bounds against the upper bound gives  $0 \leq -\varepsilon$ : a contradiction, so the full belief is nonviable. Removing mode  $q + 1$ ,  $u \equiv 1$  gives  $w_j = 1$  and  $x_j = 0$ : safe. Removing mode  $j \leq q$ ,  $u = 0$  on  $I_j$  and  $u = 1$  elsewhere gives  $x_{q+1} = (q - 1) - (q - \varepsilon) + 1 = \varepsilon \geq 0$  at the end and safety on the remaining branch: every proper subbelief is viable. Equal weights  $\lambda_j = 1/(q+1)$  on the  $q + 1$  endpoint rows give pooled pre-kernels  $\sum_j \lambda_j k_j = 0$  on every interval where the modes agree and a margin of  $\varepsilon/(q+1)$  at the horizon. The kernel rank: the endpoint rows are  $\mathbf{1}_{I_j}$ -weighted copies of the single input direction with one linear relation (their weighted sum), so the rank of the safety-row matrix is exactly  $q$ , and Helly-type sparsity in that rank requires  $q + 1$  witnesses — no bound in the input dimension  $m = 1$  exists.  $\square$

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**Derivation of the moduli from primitive data.** The hypothesis constants of Theorem 4 are not free assumptions; under the standing bounds they are computed from the data. Let  $\|A_\theta\|_\infty \leq \bar{A}$ ,  $\|B_\theta\|_\infty \leq \bar{B}$ ,  $\|E_\theta\|_\infty \leq \bar{E}$ ,  $\|g_\theta\|_\infty \leq \bar{g}$ ,  $\|c_i\| \leq \bar{c}$  on  $[0, T]$ ;  $U \subset c_U + R_U \mathbb{B}$ ,  $D_\theta \subset R_D \mathbb{B}$ ,  $P_\theta \subset R_P \mathbb{B}$ ;  $\text{TV}(B_\theta) \leq V_B$ ,  $\text{TV}(E_\theta) \leq V_E$ ; and  $M_X := e^{AT}$ , so that  $\|X_\theta(t, \sigma)\| \leq M_X$  (Grönwall on the transition matrix) and  $\frac{\partial}{\partial \sigma} X_\theta(t, \sigma) = -X_\theta(t, \sigma)A_\theta(\sigma)$ .

*Total variation of the kernels.* For  $k_a(\sigma) = \mathbf{1}_{\{\sigma \leq t\}} B_\theta(\sigma)^\top X_\theta(t, \sigma)^\top c_i$ : on  $[0, t]$  the product rule for functions of bounded variation gives

$$\text{TV}(k_a; [0, t]) \leq \sup_\sigma \|X_\theta(t, \sigma)^\top c_i\| \text{TV}(B_\theta) + \sup_\sigma \|B_\theta(\sigma)\| \int_0^t \|X_\theta(t, \sigma)A_\theta(\sigma)^\top c_i\| d\sigma \leq \bar{c}(M_X V_B + \bar{B} M_X \bar{A} T),$$

and the drop to zero at  $\sigma = t$  adds at most  $\|B_\theta(t)^\top c_i\| \leq \bar{B}\bar{c}$  (using  $X_\theta(t, t) = I$ ). Hence (7) holds with  $V_U = \bar{c}(\bar{B} + M_X V_B + \bar{B} M_X \bar{A} T)$ , and identically for the disturbance kernel with  $V_D = \bar{c}(\bar{E} + M_X V_E + \bar{E} M_X \bar{A} T)$ .

*$L^1$  label-continuity modulus.* The same bounds give the  $L^1$  continuity in the label time that the proof of (9) consumes (the sup-norm form is false here: the step kernels jump at  $\sigma = t$  for every  $t$ ). For  $t \geq t'$ , Grönwall on  $X_\theta(t, \sigma) - X_\theta(t', \sigma) = \int_{t'}^t A_\theta(r) X_\theta(r, \sigma) dr$  gives  $\|X_\theta(t, \sigma) - X_\theta(t', \sigma)\| \leq M_X \bar{A} |t - t'|$  uniformly in  $\sigma$ , so with  $k_a = \mathbf{1}_{\{\sigma \leq t\}} B_\theta(\sigma)^\top X_\theta(t, \sigma)^\top c_i$ ,

$$\|k_t - k_{t'}\|_{L^1} \leq \bar{B}\bar{c} |t - t'| + \int_0^{t'} \bar{B}\bar{c} M_X \bar{A} |t - t'| d\sigma \leq \bar{B}\bar{c} (1 + M_X \bar{A} T) |t - t'|,$$

and  $|\beta_t - \beta_{t'}| \leq L^{\text{tr}} |t - t'|$  by the trajectory Lipschitz bound below: the label map is  $L^1$ -Lipschitz in  $t$  with constant  $\bar{B}\bar{c}(1 + M_X \bar{A} T) + L^{\text{tr}}$ , which is the form the label-continuity hypothesis consumes.

*Uniform time-Lipschitz constant.* For a trajectory  $x$  of mode  $\theta$ :  $\|x(t)\| \leq M_X(R_P + T(\bar{g} + \bar{B}(R_U + \|c_U\|) + \bar{E}R_D))$  by Grönwall, and

$$\left| \frac{d}{dt} c_i^\top x(t) \right| \leq \bar{c}(\bar{A}\|x(t)\| + \bar{g} + \bar{B}(R_U + \|c_U\|) + \bar{E}R_D) \leq L^{\text{tr}},$$

with  $L^{\text{tr}} := \bar{c}\bar{A}M_X R_P + \bar{c}(1 + \bar{A}M_X T)(\bar{g} + \bar{B}(R_U + \|c_U\|) + \bar{E}R_D)$ . Each  $\mathcal{F}_{(\theta, i, t)}(\pi)$  is a supremum over  $(x_0, d)$  of the  $L^{\text{tr}}$ -Lipschitz functions  $t \mapsto c_i^\top x(t) - b_i$ , and a pointwise supremum of  $L$ -Lipschitz functions is  $L$ -Lipschitz; hence (6) holds with  $L = L^{\text{tr}}$ . For the three-branch instance the closed form applies (with  $\bar{A} = 1$ ,  $M_X = e^{6/5}$ ,  $R_P = 34/25$ ,  $\bar{B} = 1$ ,  $R_U = 1$ ); the direct kernel computation of Section 6 gives the sharper instance constants  $V_U = 6/5$ ,  $L = 11/5$ ,  $R_U = 1$ ,  $\delta_\beta = 0$  used in the sandwich reading of the campaign.

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**Proof of the erosion identity.**  $\phi_U$  is concave and positively homogeneous; for any decomposition,  $\phi_U(\sum_j g_j) \geq \sum_j \phi_U(g_j)$  because each  $\phi_U(g_j) = \min_{v \in U} g_j^\top v$  is attained at a common minimizer for the sum while separate minima only lower the total. Consequently the same witness family, evaluated on the coarse partition, dominates its evaluation on the refinement,  $\Gamma_{\text{coarse}} \geq \Gamma_{\text{fine}}$ , and the exact erosion is the displayed integral (12). For  $U = [-\bar{u}, \bar{u}]^m$ ,  $\phi_U(g) = -\bar{u}\|g\|_1$ , giving the stated  $\ell^1$  form. The 3,240-pair verification for the hexagonal  $U$  is reported in S2.  $\square$

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**Proof of the oracle-recourse certificate (13).** The oracle relaxation replaces the true post- $\tau$  policy class by the larger class that may assign each stored scenario its own measurable control. The certificate (13) is the atomic margin (1) evaluated in that enlarged class: before  $\tau$  the pooled kernel term is  $\phi_U(\sum_\ell \lambda_\ell k_\ell)$ , after  $\tau$  each scenario  $\ell$  contributes  $\phi_U(\lambda_\ell k_\ell)$  with its own margin  $\beta_\ell$  computed from the stored scenario. A positive margin certifies infeasibility in the enlarged class by Propositions 1–2; every policy of the true class is feasible for the enlarged one, so the verdict transfers downward: no actual output-feedback policy is safe. Soundness is thus immediate; completeness fails because the oracle class is strictly larger.

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## S2. The verification record

All numbers in the main text are re-derived in exact rational arithmetic (Python `fractions`; no floating point in any witness computation; floats only inside the solver call, whose output is compared against the exact value only within solver tolerance). Two scripts carry the record, both at <https://github.com/MIKEAA2020/general-sustainability> (folder `arena agent 1/paper rewrites/latex`):

- `paper2c_lp_campaign.py` — the solver-assisted, exactly certified mesh campaign: builds program (5) at each mesh  $h \in \{0.2, 0.1, 0.05, 0.02, 0.01\}$ , verifies the dimension identities ( $mQ+1$  variables;  $Ms(N_t+1) + p_U Q$  rows), solves with HiGHS, and certifies optimality of  $\rho(h) = \max\{0, \frac{3}{50} - Th/4\}$  by the exact primal witness (pre-observation blocks 0; post-observation blocks of mode  $j$  at  $-n_j$ ; every row checked) and the exact dual witness ( $\lambda = (\frac{3}{8}, \frac{5}{16}, \frac{5}{16})$ ,  $\mu_{j,k} = \lambda_j a_k \eta_j$ ; stationarity and objective  $-\rho(h)$  per block). Also re-derives  $\Gamma(\tau) = \tau - \frac{7}{50}$ ,  $\tau_{\max} = \frac{7}{50}$ , the  $h = \frac{1}{10}$  Farkas contradiction  $-\frac{3}{100}$ , and the pair-viability positions  $1.9752 = \frac{12345}{6250}$  and  $1.9352 = \frac{2419}{1250}$ , attained by the recorded pairwise policies (blind-window hold  $u = 0$  resp.  $u = n_1$ , then the two-phase brake  $-\frac{2}{5}n_j$  for  $\frac{1}{5}$  and  $-n_j$  to rest; Section 6 of the main text). The witness pair is mesh-generic on the aligned family: re-running it exactly at the sixth aligned mesh  $h = \frac{1}{15}$  ( $Q = 48$ , 687 inequalities) certifies  $\rho = \frac{1}{25}$  there, extending the law’s verified domain beyond the five campaign meshes. Current run: 9/9 checks pass; solver agreement at most  $4.2 \times 10^{-17}$  on all five meshes.
- `paper2_nonstandard_verification.py` — the 51-check exact-arithmetic record for the predecessor material: the three-branch instance values, the  $m+1$  construction for all tested  $g$ , the  $\phi_U$  superadditivity on 3,240 rational pairs, the error-bound checks, and the counterexamples. Current run: 51/51 checks pass.

The mesh table of the main text is regenerated verbatim by the first script; the reported iteration counts and wall-clock times are measured values from the run recorded here (scipy HiGHS, dual simplex), not estimates.